Closing Auction Trading: Predicting the AAPL Closing Print from L2 Order-Book Dynamics

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1. Strategy Description

Economic thesis. The strategy targets structural friction in the final hour of trading: a large share of institutional equity volume is concentrated in the closing auction, but official imbalance information is not fully available early enough for simple arbitrage. In the presentation, the key idea is that Market-on-Close and Limit-on-Close demand creates directional pressure before the auction print. That pressure can leave traces in the continuous limit-order book, including bid/ask depth changes, microprice drift, acceleration into the close, and quote cancellations. The thesis is falsifiable: if those pre-close L2 features do not predict the next 12-minute return or if the hit rate falls toward the break-even level, the strategy should be scaled down or stopped.

Trading rule. Each day, the strategy observes AAPL XNAS.ITCH MBP-10 data from 3:15 PM to 3:45 PM. It predicts the 12-minute mid-price return from 3:45 PM to 3:57 PM using a Ridge regression, enters at 3:45 PM only when the predicted move is economically large enough, and exits before the 4:00 PM closing auction. The design intentionally avoids overnight risk and avoids holding into the final auction print, where liquidity and order-flow behavior can change quickly.

Feature	Window	Interpretation
imb_early	3:15-3:35	Exponentially decayed depth imbalance; sustained early positioning.
imb_late	3:35-3:45	Late imbalance, where MOC-related flow may concentrate.
mom_30	Full 30 min	Microprice return in bps; main directional anchor.
mom_accel	Last 15 min	Acceleration: recent momentum minus full-window momentum.
cancel_imb	Full window	Ask-vs-bid cancellations; liquidity withdrawal before the auction.

Model and signal. All features are standardized using training-period means and standard deviations. Ridge regression is used instead of OLS because the training sample has only 90 days and the L2 features are correlated. The chosen penalty, alpha = 15, stabilizes coefficients while preserving the sign and magnitude of economically meaningful predictors. The fitted coefficients show that momentum and acceleration dominate the signal, while late bid depth and cancellation imbalance carry negative signs, consistent with the interpretation that dealer absorption and liquidity withdrawal can precede downward auction pressure.

Entry, sizing, and exits. At 3:45 PM, the strategy trades only if |predicted return| is at least 1.5 bps and the spread is no wider than 3 cents. It buys when the predicted return is positive and sells when it is negative, crossing the actual bid/ask rather than assuming midpoint fills. Position size is small and signal-scaled, with 1 to 5 shares in the project backtest. The exit rule is adaptive: stop out after an adverse move of 12 bps, take profit after a favorable move of 22 bps, or close the position at 3:57 PM. The take-profit/stop-loss ratio is 22/12 = 1.83, so the break-even win rate is about 35.3%, lower than a symmetric payoff strategy.

2. Backtest Summary

Data and split. The backtest uses AAPL MBP-10 order-book data from Databento XNAS.ITCH over 158 trading days, from September 27, 2024 to May 15, 2025. The temporal split is 90 training days and 68 out-of-sample days. Model weights are frozen before the out-of-sample period, and the trading threshold is set from transaction-cost economics. Transaction costs are embedded through actual bid/ask execution: buys fill at the ask, sells fill at the bid, and the strategy pays a full spread over a round trip.

Metric	In-sample	Out-of-sample
Net return (% of one-share notional)	+2.93%	+5.18%
Annualized return	+8.21% / yr	+19.19% / yr
Annualized volatility	2.22% / yr	3.80% / yr
Sharpe ratio	3.70	5.05
Win rate	54.5% (36/66)	59.5%
Max drawdown	-0.56%	-1.38%
Days traded	66 / 90	42 / 68

Percentage convention. Following the feedback, PnL is reported as a return measure rather than a raw dollar amount. The normalization uses the approximate one-share AAPL notional in the sample, about $220, so the previous $6.45 in-sample PnL is about 2.93% and the $11.39 OOS PnL is about 5.18%. This keeps the small-share proof-of-concept comparable to a scalable strategy while preserving the original Sharpe, win-rate, and execution assumptions.

Interpretation of results. The in-sample model has R-squared of 0.096 and a directional hit rate of 55.6%, which is modest but economically meaningful given the asymmetric payoff design and short holding horizon. The trading results are positive both in-sample and out-of-sample after converting PnL into a percentage return. OOS performance is stronger, with Sharpe 5.05 and a 5.18% normalized net return on the small-share test scale. The slides partly attribute this partly to the March-April 2025 tariff-driven AAPL sell-off, where realized intraday moves were larger when the signal fired. This makes the result encouraging but not conclusive: the OOS sample includes only 42 traded days, so confidence intervals remain wide.

Diagnostics and beta. The cumulative PnL plots trend upward in both samples. Rolling Sharpe is mostly above 1 in-sample and improves meaningfully out-of-sample after an early drawdown. Daily inventory stays within +/-2 shares and is flat by 3:57 PM, matching the intended intraday design. A regression of daily strategy returns on SPX daily returns shows in-sample beta near zero, with R-squared = 0.003. In the OOS period, beta becomes positive and significant, with R-squared = 0.089, reflecting a long tilt during the tariff regime. The annualized alpha estimates remain positive under the same percentage-normalized convention, but the OOS beta exposure should be treated as a disclosed risk rather than pure market-neutral alpha.

3. Risk Discussion

Regime risk. The strategy is most vulnerable when the observed 3:15-3:45 PM order-book pattern stops representing the true closing-auction pressure. A macro event, earnings headline, index rebalance, large block print, or sudden news shock between entry and exit can overwhelm the signal and trigger the stop-loss. A late MOC flip can also invalidate the prediction: a large opposing order after the observation window may reverse the imbalance, making the 3:45 PM signal stale. Thin liquidity is another failure mode because a small number of orders can distort late imbalance features and produce noisy predictions.

Capacity and transaction costs. The backtest is intentionally small scale, so the reported returns should be interpreted as proof-of-concept rather than deployable portfolio performance. Capacity breaks when expected market impact approaches the 1.5 bps entry threshold. The presentation estimates negligible impact at the current 1-5 share scale, less than 0.5 bps around a $10 million deployment, 2-4 bps around $100 million, and more than 20 bps around $1 billion. Therefore, the practical capacity is roughly $50-100 million for this single-name implementation. Scaling beyond that would require spreading orders across time, using more passive execution, trading a basket of liquid names, or accepting lower expected edge.

Signal decay and implementation risk. If other participants discover the same pre-close L2 pattern, the signal may decay through earlier price adjustment, wider spreads, or more aggressive liquidity withdrawal. The proposed decay monitor is to flag the strategy when trailing 30-day R-squared falls below 5% or when hit rate approaches the 35.3% break-even win rate; the model should be reviewed and retrained quarterly. Implementation risks include data latency, clock synchronization, venue access, queue position if passive orders are introduced, and realistic modeling of fills during the final minutes. Because the current implementation crosses the spread, it is conservative on fill price but may understate impact if scaled.

Overall assessment. The project supports the existence of an intraday closing-auction signal in AAPL over the tested period. The strongest evidence is that the strategy remains profitable after a clean temporal split, pays actual bid/ask costs, and keeps positions flat before closing. The main limitations are small OOS sample size, single-ticker scope, in-sample feature iteration, and residual OOS market beta.

Future direction. The next model upgrade should use more same-day information before the closing window. Instead of relying only on 3:15-3:45 PM L2 features, the model can add intraday trend variables such as open-to-3:15 return, VWAP deviation, realized volatility, cumulative volume, and the persistence of morning-to-afternoon momentum. This would test whether the closing-auction signal is stronger when it aligns with the broader intraday trend, and whether late L2 pressure is still useful after controlling for the day's earlier price path. A clean implementation would freeze the current Ridge framework, add these trend features in training only, and compare OOS performance against the current L2-only baseline across additional liquid Nasdaq names and different event regimes.

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